



The Identity

Artist's Proof 08

General Relativity

Einstein's field equations derived from the record algebra

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Artist's Note

This is Artist's Proof 08 of The 420 Code. The paper derives Einstein's field equations from the record algebra. It is the fourth and closing chapter of Notebook II — Spacetime, following AP03 (The Ratio), AP05 (The Break), and AP10 (The Dimension).

Not by assuming the equations.

Not by importing them from differential geometry.

By forcing them from the symmetries of the record algebra under embedding into a four-dimensional Lorentzian manifold, plus the uniqueness theorems of potential theory (weak-field limit) and Lovelock 1971 (full tensor form).

The derivation has two halves. The weak-field limit (section 4) uses four axiom-derived symmetry constraints plus the defining condition of the weak-field regime to leave a one-parameter Helmholtz family $a \cdot \nabla^2 \Phi + b \cdot \Phi = A \cdot \rho$; the Poisson equation is recovered as the small-cosmological-constant limit ($b = 0$). The full tensor form (section 8) uses five covariant constraints plus Lovelock 1971 (general N) and Lovelock 1972 (3+1 uniqueness) to force $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ as the unique symmetric, divergence-free, rank-2 tensor identity in 3+1 dimensions; the $b \cdot g_{\mu\nu}$ term is retained as the cosmological constant Λ . Both halves rest on AP10's closure of $N = 3$ (contingent on AP10's KS-D.3 remaining unfired).

The coefficient identification (section 5) carries a disclosed structural circularity: κ is defined as $G m_e^2/2$, so the relabelling $G = 2\kappa/m_e^2$ does not independently compute G . What is independently derived is the *form* of the identity (Poisson in section 4, tensor in section 8), not the numerical value of G .

Eight kill switches are tracked. KS-I.6 ($N = 3$) is CLOSED by AP10. With KS-I.6 closed, Einstein's field equations are an unconditional theorem of the record algebra in the dimensional sense — unconditional relative to the dimensional precondition, contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. KS-I.1 (axiom-to-Poisson chain, now reframed around the Helmholtz-family + small- Λ -limit recovery), KS-I.2 (Lovelock chain), KS-I.3 (coefficient identification), KS-I.4 (κ inaccessibility), KS-I.5 (curvature-record identity), KS-I.7 (cosmological constant value), and KS-I.8 ($R \rightarrow$ conservation bridge) remain live. KS-I.6's closure is contingent on AP10's

KS-D.3 remaining unfired; if KS-D.3 fires, the dimensional count changes and Lovelock's 3+1 uniqueness may not apply.

AP08 is logically independent of AP01 and AP04 in the strict sense that its postulates (the four canonical axioms, the proven hypotheses EH and QRA, and Lovelock's theorem as external mathematical input) are self-contained. AP08 *depends on* AP20 (which proves EH and QRA), AP10 (which closes $N = 3$), AP06 (Theorem 3.1 for the horizon-leakage argument in section 6), and on The Lock identification (Edition 04) of ε with the electron mass for the coefficient identification in section 5.

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Orientation

Where this paper fits in The 420 Code

The 420 Code is organised across eight Notebooks. This paper is the fourth and closing chapter of Notebook II — Spacetime. The Notebook II chapter sequence runs AP03 (The Ratio) → AP05 (The Break) → AP10 (The Dimension) → AP08 (this paper).

Notebook II derives the structure of space, time, gravity, and the dimensional count from the axiom. AP03 derives the constants. AP05 walks the chain from the axiom to obligation and reformulates the cosmological identity as junction conditions. AP10 pins $N = 3$. AP08 closes Notebook II by deriving Einstein's field equations — first the Poisson equation as the weak-field linearisation, then the full tensor form via Lovelock 1971. With AP10's closure of $N = 3$, the derivation is unconditional.

AP08 is the structural payoff of Notebook II. The bidirectional relationship with AP10 is the centre of the Notebook: AP08 derived Einstein's field equations conditional on $N = 3$; AP10 derived $N = 3$ (contingent on KS-D.3) and removed the dimensional condition; the joint output is Einstein's field equations as a theorem in the dimensional sense — unconditional relative to the dimensional precondition, with the standing contingency on AP10's KS-D.3 named throughout. Together AP10 and AP08 close the chain from four axioms to Einstein's equations with cosmological constant.

AP08 is logically independent of AP01 and AP04 in the sense that its own postulates (the four canonical axioms, EH, QRA, and Lovelock's theorem) are self-contained. AP08 depends on AP01 only for the formal apparatus of the record algebra; if AP01 stands, AP08's argument runs cleanly; if AP01 fails, the axioms fall back to declared assumptions and AP08's argument runs against those assumptions instead.

How to read this paper

The reading order is front to back. Section 0 names the dependencies and the scope. Section 1 states the four axioms and the two proven hypotheses (inherited from AP10 and AP20). Section 2 introduces the condensate — the

manifold understood as the structure that emerges from accumulated records. Section 3 defines record density on the manifold and identifies it with matter density. Section 4 derives the Poisson equation as the weak-field limit of the curvature-record identity. Section 5 identifies the coefficient and discloses the structural circularity of κ . Section 6 verifies the strong-field limit against Hawking radiation as a consistency check. Section 7 names the full nonlinear topology and the Newton-versus-Einstein limits. Section 8 derives the full tensor form via Lovelock 1971. Section 9 is the falsification register. Section 10 is the synthesis.

Two voices alternate as the material requires. Narrative passages — the condensate as the ground you stand on, the Newton-as-linearisation framing, the closing on the gravitational sector being closed without reservation — speak directly. Formal sections — the axiom statements, the symmetry constraints, the uniqueness theorems, the kill switches — speak in the precision the structure requires.

0 — Dependency and Scope

AP08 is the fourth and closing chapter of Notebook II. Before the body, this section names what AP08 does, what it depends on, where its inputs map onto the canonical axioms {S, B, R, C}, what epistemic status each section holds, what kill switches are engaged, what structural debts remain, and how AP08 stands in relation to the rest of The 420 Code.

0.1 — What this paper does

AP08 derives the form of Einstein’s field equations with cosmological constant from the four canonical axioms of the record algebra. The derivation has two parts. First, the weak-field limit: four axiom-derived symmetry constraints (Lorentz invariance, locality as differential form from Axiom C, second-order curvature from EH+QRA, scalar form of the source and the potential) applied to the weak-field regime (Constraint 2 as the defining condition of the regime, not an axiom-derived constraint) leave a one-parameter family — the Helmholtz form $a \cdot \nabla^2 \Phi + b \cdot \Phi = A \cdot \rho$ — of which pure Poisson is the small-cosmological-constant member. The Poisson equation is recovered as the leading-order static weak-field expression of section 8’s full identity, in the limit of negligible Λ corrections at laboratory scales. Second, the full tensor form: five covariant constraints (general covariance, symmetric rank-2 tensor structure, divergence-free source via the $R \rightarrow$ conservation bridge, second-order in the metric from QRA, four-dimensional manifold from AP10’s closure of $N = 3$ contingent on AP10’s KS-D.3) plus Lovelock’s 1971 theorem force $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ as the unique symmetric, divergence-free, rank-2 tensor identity in 3+1 dimensions, with the $b \cdot g_{\mu\nu}$ term retained as the cosmological constant Λ .

The contribution is the form of Einstein’s field equations as a theorem of the record algebra under the proven faithful embedding plus AP10’s closure of $N = 3$. The cosmological constant Λ is predicted to exist by Lovelock; its value is not determined by the derivation (KS-I.7). The coefficient is identified — not derived — as $8\pi G/c^4$ via The Lock identification $G = 2\kappa/m_e^2$ (section 5); the structural circularity is honestly disclosed and tracked by KS-I.3 and KS-I.4.

AP08 also verifies the strong-field limit against established physics (section 6) by recovering the Hawking temperature from the Schwarzschild solution of section 8’s field equations plus AP06 Theorem 3.1 (nonzero leakage when c is

finite). The numerical formula $T_H = \hbar c^3 / (8\pi k_B G M)$ is the standard Hawking 1975 result; what the 420 Code contributes is the structural interpretation (the event horizon as the locus where the fold becomes total) plus the structural guarantee of nonzero leakage. The Hawking temperature written as $T_H = \hbar c^3 m_e^2 / (16\pi k_B G M)$ after substituting $G = 2\kappa / m_e^2$ is the same Lock relabelling from section 5.1 reappearing at the horizon scale, not an independent structural finding about the electron at the horizon.

0.2 — Dependencies

AP08 is a derivation paper with dependencies on both proven internal results and one external mathematical theorem (Lovelock 1971).

AP20 (the canonical axioms). Load-bearing for the four canonical axioms $\{S, B, R, C\}$ (formal statement at AP20 section 3.3; completeness and minimality stated at AP20 section 4 with proof sketches, fenced at KS-P.1/KS-P.2; joint consistency exhibited by any existing record — the undeniable premise, AP20 section 6). The axioms enter section 1 as the foundation; their independence is invoked indirectly via AP10’s closure of $N = 3$.

The Lorentzian signature (AP10 section 4, fenced at KS-D.4). Load-bearing for the Lorentzian manifold (M, g) with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$, and for special relativity as a theorem (not an assumption). The signature is carried by AP10 section 4’s structural argument (R’s unique irreversibility reads as the $(-)$ metric character), with the formal metric-level derivation fenced at AP10’s KS-D.4. The signature is the precondition for Constraint 1 of section 4 (Lorentz invariance) and Constraint 1 of section 8 (general covariance).

AP20 (EH and QRA proven). Load-bearing for the faithful embedding of the record algebra into a smooth manifold (EH) and for the differentiability and leading-order quadratic structure of the cone boundary (QRA). EH underwrites the entire derivation — the record algebra must embed into a smooth manifold for the curvature equations to be meaningful. QRA is invoked in Constraint 4 of sections 4 and 8 (second-order curvature requires the metric to be at least twice-differentiable).

AP10 (The Dimension). Load-bearing for $N = 3$ in Constraint 5 of section 8. AP10 closes KS-I.6; with $N = 3$ derived from the axioms, Lovelock’s theorem applies in 3+1 dimensions and the uniqueness of $G_{\mu\nu} + \Lambda g_{\mu\nu}$ as the

divergence-free symmetric rank-2 tensor is forced. The closure of KS-I.6 is contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired; if KS-D.3 fires, the dimensional count changes and Lovelock's 3+1 uniqueness may not apply.

AP06 Theorem 3.1 ($\varepsilon > 0$ whenever c is finite). Load-bearing for the strong-field limit at section 6.2. The Hawking-radiation argument requires the horizon to leak nonzero radiation; AP06 Theorem 3.1 establishes that the leakage is structurally guaranteed by the finiteness of c .

The Lock, Edition 04 ($\varepsilon = \text{electron}$). Load-bearing for the coefficient identification at section 5. The relabelling $G = 2\kappa/m_e^2$ — and therefore the writing of the coupling $8\pi G/c^4$ as $8\pi\kappa/(c^4 m_e^2)$ — depends on identifying ε with the electron. KS-I.3 is the structural test on this identification.

Lovelock 1971 (theorem). External mathematical input. The proof that in N spacetime dimensions, the only symmetric, divergence-free, rank-2 tensor constructed from the metric and its first and second derivatives takes a specific form. In 3+1 specifically (Lovelock 1972) the form is uniquely $G_{\mu\nu} + \Lambda g_{\mu\nu}$. The theorem is mathematics, not internal to the body of work; AP08 invokes it as such.

0.3 — Axiom mapping

AP08 inherits the four canonical axioms from AP20 Paper D and AP10, and uses them to constrain the form of the curvature-record identity. The mapping is direct rather than retrospective.

Axiom S (Symmetry). Two disjoint sectors $\mathcal{L}$ and $\mathcal{P}$ with order-reversing involution σ . Underwrites the eye topology of section 7 (two curves meeting at the common point) and the structural framing of the condensate as the manifold that emerges from the symmetric/asymmetric structure.

Axiom B (Unique Breaking). One element $\varepsilon \in \mathcal{L}$ with no σ -image; $\nu(\varepsilon) = 1$. Under The Lock identification, $\varepsilon = \text{electron}$. Underwrites the matter-density identification at section 3.2 (every record involves an electron coupling event), the coefficient identification at section 5 ($G = 2\kappa/m_e^2$), and the strong-field leakage at section 6.3 (the electron appears at the horizon).

Axiom R (Record Monotonicity). Monoid, not group; history is irreversible. Underwrites the linearity-in-weak-field constraint of section 4 (records accumulate additively, leading to a linear leading-order relationship between curvature and density), the divergence-free source in section 8 (the $R \rightarrow \nabla^\mu T_{\mu\nu} = 0$ bridge, KS-I.8), and the non-negativity of the density field ($\rho \geq 0$ everywhere) at section 3.3.

Axiom C (Constraint). Finite invariant rate c . Underwrites the locality constraint of section 4 (curvature at x depends only on density at x and its immediate neighbourhood), the propagation bound on density changes (ρ propagates at most at c), and the horizon-formation argument at section 6.1 (when the fold becomes total, C creates a one-way boundary).

0.4 — Epistemic status per section

Each section carries an explicit status — derivation, identification, formal block, or synthesis — so the reader knows what kind of weight each carries.

Section 1 (Axioms Inherited). FOUNDATIONAL. The four axioms re-stated from AP20 Paper D; EH and QRA named as proven (AP20); the Lorentzian manifold with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$ carried via AP10 section 4 (fenced at KS-D.4); special relativity as a theorem of that signature.

Section 2 (The Condensate). STRUCTURAL FRAMING. The manifold understood as accumulated record structure (not as a pre-existing container). Mode 0 (the metric) and Mode 1 (propagation events) named.

Section 3 (Record Density). DEFINITION + IDENTIFICATION. The record number density $n(x)$ defined; the identification with matter density $\rho(x) = n(x) \cdot \langle E \rangle / c^2$ made under The Lock; properties inherited from axioms ($\rho \geq 0$ from R , causality from C , electron coupling from B).

Section 4 (Form of the Identity — Weak-Field Limit). DERIVATION + LIMIT RECOVERY. Four axiom-derived symmetry constraints plus the defining condition of the weak-field regime leave a one-parameter Helmholtz family; pure Poisson is recovered as the small-cosmological-constant member of that family. KS-I.1 is the structural test on the family identification and the small- Λ recovery.

Section 5 (The Coefficient). IDENTIFICATION + HONEST DISCLOSURE. The coefficient $A = 4\pi G$ identified via The Lock as $8\pi\kappa/m_e^2$. Structural circularity (κ defined as $Gm_e^2/2$) explicitly disclosed. KS-I.3 (coefficient identification) and KS-I.4 (κ inaccessibility) are the structural tests.

Section 6 (Strong-Field Limit — Hawking Verification). CONSISTENCY CHECK — RECOVERY. The standard Hawking temperature recovered from the Schwarzschild solution of section 8's field equations plus established quantum field theory in curved spacetime (Hawking 1975); the condensate picture and AP06 Theorem 3.1 supply the structural interpretation and the nonzero-leakage guarantee. The numerical formulas are imported; the structural picture is the body of work contribution. The m_e in T_H (equation 6.3) is the same Lock relabelling from section 5.1 reappearing at the horizon scale, not an independent finding.

Section 7 (Full Nonlinear Structure). STRUCTURAL ARGUMENT. The eye topology, Mode 0 / Mode 1, Newton as linearisation, Einstein as full curve.

Section 8 (Tensor Form). DERIVATION. Five covariant constraints plus Lovelock's 1971 theorem force $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. KS-I.2 is the structural test on the uniqueness of Lovelock. KS-I.6 ($N = 3$) is closed by AP10, contingent on AP10's KS-D.3 remaining unfired.

Section 9 (Kill Switches). FORMAL FALSIFICATION REGISTER. Eight kill switches engaged; one (KS-I.6) closed by AP10; seven live.

Section 10 (Synthesis). What this paper closes, what it leaves open.

0.5 — Kill switch summary

Eight kill switches are engaged in this paper. One is closed by AP10; seven remain live. The full register is in section 9.

KS-I.1: Axiom-to-Poisson chain. LIVE — HARD. If the five symmetry constraints in section 4 do not uniquely determine the Poisson equation, the weak-field derivation fails.

KS-I.2: Lovelock chain. LIVE — HARD. If the five covariant constraints in section 8 do not uniquely determine the Einstein tensor plus cosmological

constant in 3+1 dimensions, the tensor derivation fails. Lovelock 1971 (general N) and Lovelock 1972 (3+1 uniqueness) are the load-bearing external claims.

KS-I.3: Coefficient identification. LIVE — HARD. If $G \neq 2\kappa/m_e^2$ — if The Lock identification of ϵ with the electron is structurally wrong — the coefficient identification in section 5 fails. The form survives; the coefficient does not.

KS-I.4: κ inaccessibility. LIVE — HARD. If an observable is found that constrains κ without going through G, the structural inaccessibility claim of section 5.2 is falsified and κ becomes independently testable.

KS-I.5: Curvature-record identity. LIVE — EMPIRICAL. If gravity is shown to source from a quantity structurally distinct from accumulated records (matter without record-history, or records without gravitational signature), the identity fails.

KS-I.6: $N = 3$. CLOSED by AP10 — contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. With $N = 3$ derived in AP10, the Lovelock derivation in 3+1 is unconditional in the dimensional sense; if KS-D.3 fires at any axiom, the count changes and Lovelock's 3+1 uniqueness may not apply.

KS-I.7: Cosmological constant value. LIVE — EMPIRICAL. The existence of Λ is predicted by Lovelock. The value is not determined. If Λ is measured to a value parametrically incompatible with the architecture's structural constraints (wrong sign, wrong magnitude class), the derivation's reach is bounded but the form survives.

KS-I.8: $R \rightarrow$ conservation bridge. LIVE — HARD. The map from discrete record monotonicity (Axiom R) to the covariant local conservation $\nabla^\mu T_{\mu\nu} = 0$ is a bridge step in Constraint 3 of section 8.1, not a completed derivation. If the bridge fails, the tensor form chain fails at Constraint 3.

0.6 — Structural debts owed

Two structural-class items are tied to AP08's argument as the 420 Code debts. Named here so the structural exposure is visible at the door.

The Lock identification ($\epsilon = \text{electron}$). Used at section 3.2 ($\rho = n \cdot \langle E \rangle / c^2$ under the electron interpretation), section 5 (the coefficient identification $G = 2\kappa/m_e^2$), and section 6.3 (the m_e in T_H). The identification is structural

rather than empirical — it is the 420 Code claim that the unique ε of Axiom B is the same ε that appears empirically as the electron — and is tracked by KS-I.3 as a live structural test. The form of Einstein’s field equations survives if The Lock fails; the coefficient does not.

The $R \rightarrow \nabla^\mu T_{\mu\nu} = 0$ bridge. Constraint 3 of section 8.1 requires the source tensor to satisfy covariant local conservation. The motivation comes from Axiom R (records cannot be created or destroyed locally), but the precise map from discrete record monotonicity to the covariant conservation law in the continuum limit is a bridge step, not a completed derivation. KS-I.8 is the live structural test on this bridge. If the bridge fails, the tensor derivation fails at Constraint 3; the weak-field expression survives (it does not invoke covariant conservation).

0.6.1 — External mathematical dependencies

Separately from the 420 Code structural debts, AP08 invokes external mathematical results as established theorems. These are not the 420 Code debts — the body of work does not attempt to derive them — but they are tracked at structural-test level so the chain is auditable end-to-end.

Lovelock 1971 and Lovelock 1972. The uniqueness theorems for divergence-free symmetric rank-2 tensors in general N (Lovelock 1971) and specifically in $3+1$ dimensions (Lovelock 1972). Invoked at section 8.2. KS-I.2 is the structural test on whether the theorems hold as invoked.

Potential theory in the Helmholtz-family identification. Section 4.2 invokes the standard fact that the most general local, linear, second-order, rotationally-invariant scalar operator acting on a scalar field is $a \cdot \nabla^2 + b \cdot (\text{identity})$. KS-I.1 is the structural test on this identification and on the small- Λ recovery of pure Poisson.

The Schwarzschild solution and Hawking 1975. Section 6 recovers the standard Hawking temperature from the Schwarzschild solution of section 8’s equations plus the established quantum-field-theory-in-curved-spacetime derivation. The numerical formulas are external; the structural picture is the 420 Code contribution.

0.7 — Items held open without claiming debt status

Two items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as the items in 0.6.

The cosmological constant value (KS-I.7). Lovelock’s theorem predicts the existence of Λ ; the value is not determined by the derivation. Whether Λ can be computed from a downstream architectural principle, or whether it remains an empirical input, is open work — held open here without claiming a separate structural debt because the existence claim is what AP08 establishes; the value is downstream cosmological work.

Constraint 2 of section 4 (the weak-field regime as defining condition).

The weak-field regime is the regime section 4 derives in; the linearity going into the Helmholtz family is the defining condition of that regime, not an axiom-derived constraint. The full nonlinear extension (section 7, section 8) does not require it; Lovelock’s theorem works independently of linearity. Held open here without claiming a separate debt because “weak-field” is a chosen regime, not an unresolved structural item.

0.8 — Structural relationships

AP08 is the closing chapter of Notebook II. The chapter sequence runs AP03 → AP05 → AP10 → AP08. AP03 derives the constants (c , G); AP05 walks the chain from axiom to obligation and reformulates the cosmological identity as junction conditions; AP10 pins $N = 3$ and identifies the Hilbert space as the 1:1; AP08 closes with Einstein’s field equations.

AP01 (Notebook I) is the source of the record algebra. AP20 (Notebook I) is the source of EH, QRA, and the canonical four axioms (formal statement at AP20 section 3.3; completeness and minimality fenced at KS-P.1/KS-P.2). AP10 (Notebook II) closes the $N = 3$ conditional that AP08’s tensor derivation requires. AP06 (Notebook IV) provides Theorem 3.1 (leakage when c is finite) used in section 6. The Lock (Edition 04) provides the ε = electron identification used in section 5.

Downstream, AP08 feeds the entire gravitational and cosmological programme. Predictions $\emptyset$ inherits the unconditional Einstein’s field equations and the predicted existence of Λ . Dissolutions $\emptyset$ inherits the philosophical-register reading of gravity as record-density curvature. The horizon-leakage result at

section 6 feeds AP24 (the residual programme) and the strong-field limit of cosmological models.

1 — Axioms (Inherited)

The four axioms of the record algebra, formally stated at AP20 section 3.3 and walked in detail in AP10 section 1.

S (Symmetry). Two disjoint sectors $\mathcal{L}$ and $\mathcal{P}$ with order-reversing involution σ . $Q(\mathcal{L}) = Q(\mathcal{P})$ in the unbroken case.

B (Unique Breaking). One element $\varepsilon \in \mathcal{L}$ with no σ -image. $\nu(\varepsilon) = 1$. The break. Under The Lock: $\varepsilon = \text{electron}$.

R (Record Monotonicity). Monoid, not group, in each sector. No non-identity element has an inverse. History is irreversible.

C (Constraint). Finite invariant rate c constraining sequential propagation. Structural, not electromagnetic.

Independent (the removal arguments at AP20 section 4.2, fenced at KS-P.2) and jointly exhibited by any existing record (AP20 section 6).

1.1 — Bridge hypotheses (now theorems)

EH (Embedding Hypothesis). In the large- N limit, the discrete record algebra admits faithful embedding into a smooth manifold M .

QRA (Quantum-Record Alignment). Quantum states are pre-state records (closed at AP20 section 5.5, KS-P.4). The cone boundary's differentiability and leading-order quadratic structure used here rest on the smooth-manifold structure fenced at KS-P.5.

Both proven in AP20.

Under EH + QRA, the record algebra produces a Lorentzian manifold (M, g) with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$.

The embedding is proven at AP20 (EH); the Lorentzian signature is carried by AP10 section 4's structural argument, with the formal metric-level derivation fenced at KS-D.4.

Special relativity is a theorem, not an assumption.

2 — The Condensate

Before the break, the system is the 1:1.

Axiom S without B. Two sectors, perfectly paired, net content zero.

The break creates ε — one unpaired element.

Under EH, the large-N limit of accumulated records produces the smooth manifold (M, g) .

The manifold IS the condensate.

Not a pre-existing container into which records are placed. The structure that emerges from the records' accumulation.

The condensate and the records are co-constitutive. Without the condensate, records have nowhere to embed. Without records, the condensate has no content.

The metric g on M encodes the condensate's geometry.

Where the metric is flat, the condensate is unstressed.

Where it curves, the condensate is folded.

The curvature is determined by its own content.

You are standing on the condensate now.

The ground beneath your feet, the air around you, the light in your eyes — all of it is accumulated records embedded on the manifold that their accumulation created.

The geometry you walk through is the geometry of what has happened.

2.1 — Mode 0 and Mode 1

The condensate admits two modes of description.

Mode 0 is the metric g itself. The geometry. The background condition. The equilibrium structure that emerges from the cumulative records as a stable background.

Mode 1 is propagation on the condensate. Coupling events. The electron interacting with the electromagnetic field. The perturbative excitations of individual record-writing events on the background.

Mode 1 requires Mode 0.

In the language of the axioms: Mode 0 is the structure that EH + QRA extract from the cumulative record algebra. Mode 1 is the individual record-writing events that continue to occur on that structure. The terminology mirrors the cumulative-vs-perturbative distinction common in condensed-matter physics; in The 420 Code, gravity lives in Mode 0 and quantum mechanics in Mode 1 (the distinction is taken up in AP09).

Gravity — the curvature of the condensate — is Mode 0.

3 — Record Density on the Manifold

You have a manifold.

You have records.

The question is: how many records are there at each point?

The answer is a number. That number is the density.

And that density — as you are about to see — is indistinguishable from what physics calls “matter.”

3.1 — The definition

Under EH, each record r in the algebra embeds as an event $e(r)$ on the manifold M .

The embedding preserves causal order (Axiom R) and the rate bound (Axiom C).

Define the record number density:

$$n(x) = \lim_{V \rightarrow 0} N(V) / |V| \quad (3.1)$$

where $N(V)$ is the number of embedded records in a spacetime volume V containing the point x , and $|V|$ is the proper volume.

A scalar field on M .

3.2 — Record density is matter density

Epistemic status: identification, conditional on The Lock (ε = electron).

Every record is an ε -event (Axiom B). Records emerge through the break, and Axiom B specifies ε as the unique element responsible for the break; every record formation event therefore involves ε by construction. The full structural argument for record-formation-via- ε lives in AP01; AP08 inherits the result. The empirical handle on the identity (whether all matter sources from accumulated records) is KS-I.5.

Under The Lock's identification $\varepsilon = \text{electron}$, every record involves an electron coupling event. Every coupling event involves energy exchange.

The energy associated with accumulated records at a point x is not a separate quantity from the energy density of matter at x .

The mass density $\rho(x)$ and the record density $n(x)$ are related by:

$$\rho(\mathbf{x}) = n(\mathbf{x}) \cdot \langle E \rangle / c^2 \quad (3.2)$$

where $\langle E \rangle$ is the average energy per record.

A statement of identity.

The matter density and the record density are two measurements of the same quantity, in different units.

3.3 — Properties inherited from the axioms

From R (Record Monotonicity). $\rho(x) \geq 0$ everywhere. Records accumulate; they do not annihilate. The density cannot go negative.

From C (Constraint). Changes in ρ propagate at most at speed c . The density field respects causality.

From B (Unique Breaking). Every record involves ε . Under The Lock's identification, the density is denominated in electron coupling events.

4 — The Form of the Identity (Weak-Field Limit)

You have a manifold.

You have a density field on it.

Now ask: what is the most general equation relating the curvature of the condensate to the record density?

You are not choosing the answer.

You are watching the constraints eliminate every alternative until only one remains.

4.1 — The constraints

Four axiom-derived constraints, plus one regime-defining condition (Constraint 2). The four axiom-derived constraints together with the regime definition leave a one-parameter family of equations; the Poisson form is recovered as the small-cosmological-constant limit (see section 4.2 and the reconciliation with section 8).

Constraint 1 (Lorentz invariance). The identity must be covariant under $SO(1, N)$. Derived from the axioms via the Lorentzian signature carried at AP10 section 4 (formal derivation fenced at KS-D.4); $SO(1, N)$ is the isometry group of that signature.

Constraint 2 (Weak-field regime — defining condition). The weak-field regime is the regime in which the metric is written as $g \approx \eta + h$ with h small, and the leading-order approximation to the curvature is linear in the perturbation. This is what “weak-field” *means*; it is not derived from Axiom R but is the defining condition of the regime section 4 derives in. The full nonlinear case (section 8) does not assume linearity; Lovelock’s theorem works independently of it. Axiom R’s record additivity is consistent with linearity in this regime but does not force it as an independent constraint.

Constraint 3 (Differential form). The identity is a local differential equation rather than an integral equation — appropriate to a finite-causal-bound

substrate (Axiom C) in which the dynamic field equations must reduce to first- and-second-derivative relations on the metric. In the static limit considered here, the differential form yields equations whose Green-function solutions can themselves be non-local (the static Poisson kernel is $1/|x - y|$); the locality is in the differential form, not in the solutions. The full causal structure is restored in the dynamic case at section 8.

Constraint 4 (Second order). Curvature is a second-derivative quantity. On the manifold produced by EH + QRA, the curvature involves second derivatives of g . In the weak-field limit where $g \approx \eta + h$, the curvature reduces to second spatial derivatives of the perturbation.

Constraint 5 (Scalar). The record density $\rho(x)$ is a scalar field. In the static weak-field limit, the gravitational potential Φ is also a scalar. The identity relates two scalar quantities.

4.2 — The form: Poisson as the leading-order limit

The four axiom-derived constraints applied to the weak-field regime determine the form of the identity up to a one-parameter family.

The most general local, linear, second-order, rotationally-invariant scalar identity on a flat background relating a potential Φ to a source ρ is:

$$\mathbf{a} \cdot \nabla^2 \Phi(\mathbf{x}) + \mathbf{b} \cdot \Phi(\mathbf{x}) = \mathbf{A} \cdot \rho(\mathbf{x}) \quad (4.1)$$

where a and b are constants, A is the coupling, and $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is the Laplacian. The general form is the Helmholtz family; pure Poisson is the $b = 0$ case.

Honest disclosure. The four axiom-derived constraints together with the regime definition do not by themselves force $b = 0$. To recover pure Poisson (the form Newton found), an additional input is required: the absence of an intrinsic length scale in Mode 0 at the regime where section 4 derives the form, equivalently the absence of Yukawa-type screening of the gravitational potential at laboratory distances. The body of work does not derive this absence as a theorem; it is recovered as the small-cosmological-constant limit of section 8's full tensor form, in which the Λ contribution becomes negligible at the scales where section 4 applies.

Reconciliation with section 8. The full tensor form $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ (section 8.3, equation 8.1) retains the b-analogue: the $\Lambda g_{\mu\nu}$ term is the four-dimensional covariant generalisation of the $b \cdot \Phi$ contribution. In the Newtonian limit of section 8's equation, the Λ term contributes a constant background offset to the equation for Φ (rather than a Yukawa-type screening), and the leading-order static behaviour at laboratory scales is the pure Poisson form. Pure Poisson is therefore the leading-order leading-scale limit of section 8's full equation, valid when corrections of order $\Lambda/(\text{laboratory scale})^2$ are negligible. Section 4's "uniqueness" is a uniqueness-up-to-the-Helmholtz-family, with the pure Poisson form recovered as the limit just described.

In that limit:

$$\nabla^2 \Phi(\mathbf{x}) = \mathbf{A} \cdot \boldsymbol{\rho}(\mathbf{x}) \quad (4.2)$$

The Poisson equation.

Not derived by assuming Newton's law or Einstein's equations. Recovered as the leading-order static weak-field expression of the full identity that section 8 derives, in the limit of negligible Λ corrections at laboratory scales.

You have just watched gravity's leading-order equation emerge from four axiom-derived constraints, the weak-field regime definition, and the small-cosmological-constant limit of the full tensor identity.

The constraints leave a one-parameter family. The Helmholtz family. The small- Λ limit picks out the Poisson member of that family. The equation Newton found by observation, recovered here from the structure plus a defining condition of the regime plus a limit.

The structural circularity is preserved: A is the coupling, identified (not derived) at section 5 as $4\pi G = 8\pi\kappa/m_e^2$. KS-I.1 is the structural test on the leading-order-family identification and the recovery of pure Poisson.

5 — The Coefficient

Equation (4.2) has the form $\nabla^2\Phi = A\rho$.

The constant A must be identified.

Through The Lock:

$$G = 2\kappa / m_e^2 \quad (5.1)$$

Therefore:

$$A = 4\pi G = 8\pi\kappa / m_e^2 \quad (5.2)$$

and the identity reads:

$$\nabla^2\Phi = (8\pi\kappa / m_e^2) \cdot \rho \quad (5.3)$$

The coefficient is the ratio of the fabric's holding limit to the square of the electron mass.

The entire gravitational sector is determined by two quantities.

How much the fabric could hold (κ).

How much had to leave (m_e).

You are looking at gravity expressed as a ratio of what stayed to what escaped.

5.1 — The circularity and why it is structural

Honest disclosure.

κ is defined as $Gm_e^2/2$.

Therefore writing $G = 2\kappa/m_e^2$ is, at the level of numerical computation, a tautology: $G = 2(Gm_e^2/2)/m_e^2 = G$.

The coefficient has been relabelled, not independently derived.

Anyone who reads section 5 and concludes that G has been computed from first principles without empirical input is mistaken.

What HAS been done.

The form $\nabla^2\Phi = A\rho$ is derived from the axioms.

The five symmetry constraints in section 4 force the Laplacian, force linearity, and leave exactly one free constant A .

The form is genuinely new.

The constant A is then identified — not derived — as $4\pi G$, or equivalently $8\pi\kappa/m_e e^2$.

5.2 — Why κ cannot be independently measured

κ is the holding limit of the pre-state — the maximum coherence the system could sustain before the break became necessary.

Measurement itself is a product of the break that κ describes.

Before the break (Axiom B), there are no records (Axiom R has nothing to order), no observers, no instruments, and no manifold (EH has nothing to embed).

The only access to κ is through its consequences — G and m_e — which are the residuals of the break.

κ is a real structural quantity with well-defined dimensions and physical meaning, but it is accessible only through its products.

Kill switch KS-I.4. If an observable is found that constrains κ without going through G , the structural inaccessibility claim is falsified and κ becomes independently testable.

You cannot measure what existed before measurement existed.

κ is the capacity of the pre-state — and the pre-state is, by definition, the state before records.

The circularity is not a bug.

It is the structure telling you what it can and cannot show you.

You are inside the break. You cannot look behind it.

6 — The Strong-Field Limit (Hawking Verification)

A consistency check against established physics.

Note on what this section does and does not do. The Schwarzschild surface gravity $\kappa_H = c^4/(4GM)$ and the Hawking temperature $T_H = \hbar c^3/(8\pi k_B GM)$ are standard results of general relativity plus quantum field theory in curved spacetime (Hawking 1975). They are recovered here from the Schwarzschild solution of section 8's field equations, not independently derived from the condensate picture. What the condensate picture and AP06 Theorem 3.1 contribute is the structural interpretation: the event horizon as the locus where the fold becomes total, and the structural guarantee that the horizon leaks nonzero radiation whenever c is finite. The numerical formulas come from established physics; the structural picture is the 420 Code contribution. (Section 6.2 forward-references section 8's field equations; the dependence is acyclic because section 8 does not invoke section 6.)

6.1 — Maximum curvature

As record density increases, the condensate folds more tightly.

When the fold becomes total, the constraint (Axiom C) creates a one-way boundary.

Records can enter but not exit.

Outward propagation is forbidden.

The event-horizon interpretation from the condensate picture.

6.2 — The leakage at the horizon

AP06 Theorem 3.1: $\varepsilon > 0$ whenever c is finite.

At the event horizon, absorption is near-perfect but not perfect.

Hawking radiation provides the established nonzero horizon-leakage channel.

The surface gravity of a Schwarzschild horizon of mass M (from the Schwarzschild solution of section 8's field equations) is:

$$\kappa_H = c^4 / (4GM) \quad (6.1)$$

The Hawking temperature follows from standard quantum field theory in curved spacetime (Hawking 1975):

$$T_H = \hbar \kappa_H / (2\pi c k_B) = \hbar c^3 / (8\pi k_B G M) \quad (6.2)$$

The Hawking temperature.

Recovered from established physics. What the body of work contributes is structural: (i) the event-horizon interpretation as the locus where the fold becomes total, (ii) the structural guarantee that the leakage is nonzero (AP06 Theorem 3.1), and (iii) the surface gravity being set by the fold's geometry. The numerical formulas are not new; the structural picture under which they are recovered is.

6.3 — The electron at the boundary (an honest note)

Substituting $G = 2\kappa/m_e$ into equation (6.2):

$$T_H = \hbar c^3 m_e / (16\pi k_B \kappa M) \quad (6.3)$$

The Hawking temperature contains the electron mass after the substitution.

Honest disclosure. The standard Hawking temperature (equation 6.2) does not contain the electron mass; it appears in equation (6.3) only because G has been rewritten as $2\kappa/m_e$ — the same relabelling disclosed in section 5.1 as a tautology at the level of numerical computation. The “appearance” of m_e at the horizon is the same identification reappearing in the strong-field expression, not an independent structural result. The honest reading: under The Lock's identification, the gravitational coupling at *all* scales involves m_e (because $G = 2\kappa/m_e$), so m_e is present wherever the coupling enters — quantum scale, laboratory scale, horizon scale. The horizon expression makes the dependence visible because it isolates the gravitational coupling in a single dimensionful combination; it does not reveal a new structural role for the electron beyond what The Lock already says.

What can be said honestly. The Lock's identification ($\varepsilon = \text{electron}$) is structurally consistent at the horizon scale as it is at the quantum scale; nothing about the strong-field limit contradicts the identification. This is consistency, not corroboration: the same identification entered both expressions, so the same identification appears in both.

7 — The Full Nonlinear Structure

The Poisson equation (section 4) is a linearisation.

It describes the weak-field regime where the relationship between curvature and record density is approximately linear.

The full relationship is nonlinear.

The curvature depends on the density, and the density depends on the curvature.

7.1 — The topology

The eye topology describes the full nonlinear curve.

Two curves (the two sectors $\mathcal{L}$ and $\mathcal{P}$ from Axiom S, related by the involution σ) meet at a common point.

The point is 0 — the 1:1, the pre-state of perfect symmetry before the break.

The topology is closed. When folded onto a sphere, the two apparent endpoints are the same point.

7.2 — Reading the structure

At the common point (0). The two sectors meet. No separation, no break, no curvature, no records. The Poisson equation is the linearisation of the full curve near this point.

Moving outward. The break occurs. ε appears. Records accumulate. The curves separate. Curvature increases.

At maximum separation. Maximum curvature. Maximum record density. The event horizon. The extremal point relative to which the full structure is measured — relativistic in the original sense (relative to its own extremal point, not to a fixed background).

The closed topology. The quantum scale (near the common point, where ε is the break) and the cosmological scale (the full extent of the curve) are

connected in the topology. The Lock's identification of ε with the electron carries through the closure: the same electron mass that defines the gravitational coupling ($G = 2\kappa/m_e^2$, section 5) appears in the horizon expression at section 6.3 — because the same identification entered both. The electron is at both ends of the gravitational identity as a structural consequence of The Lock, not as an independent finding at the horizon.

7.3 — Newton vs Einstein

Newton's gravity is the linearisation near the common point.

Curvature proportional to density, constant coefficient, valid where the fold is gentle.

Einstein's gravity is the full curve.

The geometry and the distribution of energy-momentum locked together by a nonlinear identity.

The architecture derives both.

The linearisation is forced by symmetry constraints (section 4). The full tensor form is forced by Lovelock's theorem (section 8).

You have now seen the full shape.

Newton saw the straight line near the origin.

Einstein saw the curve.

They are the same structure read at different distances from the common point.

The mathematics did not choose between them. It produced both.

You do not choose either. You inherit both.

8 — The Tensor Form

Same method as section 4, applied at the covariant level.

8.1 — The constraints

Constraint 1 (General covariance). The identity must hold in all coordinate systems. Under EH, the manifold has no preferred coordinates. A tensor equation.

Constraint 2 (Symmetric rank-2 tensor). The source is the stress-energy tensor $T_{\mu\nu}$ — the covariant generalisation of ρ . The geometry side must match: symmetric rank-2 tensor built from the metric.

Constraint 3 (Divergence-free source; bridge compatibility). To write a consistent local covariant field identity with matter coupling on the embedded manifold, the source tensor is required to satisfy $\nabla^\mu T_{\mu\nu} = 0$. Motivated by Axiom R: because records cannot be created or destroyed locally, the total number of records in any region changes only due to flow across the boundary, and in the continuum limit this conservation law translates to the vanishing of the covariant divergence of the stress-energy tensor. The exact map from record monotonicity to covariant local conservation is a bridge step, not a completed derivation in this paper (KS-I.8).

Constraint 4 (Second-order in the metric). The curvature involves second derivatives of g (from QRA). The geometric tensor must be built from $g_{\mu\nu}$ and its first and second derivatives.

Constraint 5 (Four-dimensional). The manifold has dimension $3+1$. $N = 3$ is derived from the four independent axioms (AP10, The Dimension). KS-I.6 is CLOSED — contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. With $N = 3$ derived under that contingency, the dimensional precondition for Lovelock's $3+1$ uniqueness theorem is met.

8.2 — Lovelock's theorem

Theorem (Lovelock, 1971; uniqueness in $3+1$ specifically, Lovelock 1972). In four spacetime dimensions, the only symmetric, divergence-free,

rank-2 tensor constructed from the metric tensor and its first and second derivatives is:

$$\mathbf{a} \cdot \mathbf{G}_{\mu\nu} + \mathbf{b} \cdot \mathbf{g}_{\mu\nu}$$

where $\mathbf{G}_{\mu\nu} = \mathbf{R}_{\mu\nu} - \frac{1}{2}\mathbf{R}\mathbf{g}_{\mu\nu}$ is the Einstein tensor, $\mathbf{g}_{\mu\nu}$ is the metric tensor, and $\mathbf{a}$, $\mathbf{b}$ are constants.

The tensorial analogue of the uniqueness of the Laplacian.

The proof is constructive and independent of physical content — pure differential geometry. Lovelock 1971 derives the result in general N ; Lovelock 1972 establishes the special uniqueness of 3+1 from which the form above follows. AP08 invokes both as external mathematical input, tracked by KS-I.2.

8.3 — The identity in tensor form

Constraints 1–5 from the axioms, plus Lovelock’s theorem, force the full identity:

$$\mathbf{G}_{\mu\nu} + \Lambda \mathbf{g}_{\mu\nu} = (8\pi\mathbf{G}/c^4) \mathbf{T}_{\mu\nu} \quad (8.1)$$

$\mathbf{G}_{\mu\nu}$ is the Einstein tensor — the unique divergence-free, symmetric, rank-2 curvature tensor in four dimensions.

$\mathbf{T}_{\mu\nu}$ is the stress-energy tensor — the covariant record density.

Λ is the cosmological constant. Predicted to exist by Lovelock’s theorem (allowed by the constraints). Its value is not determined by the derivation (KS-I.7).

$8\pi\mathbf{G}/c^4$ is the coupling. The $\mathbf{G}$ inside is the same $\mathbf{G}$ identified in section 5 through The Lock.

Einstein’s field equations with cosmological constant.

Derived from four axioms, symmetry constraints, EH + QRA, and Lovelock’s theorem.

With AP10’s closure of $N = 3$ (contingent on KS-D.3 remaining unfired), the derivation is unconditional in the dimensional sense — unconditional relative to

the dimensional precondition, with the standing contingency on AP10's KS-D.3 named explicitly throughout.

The cosmological constant is not added by hand. Its existence is predicted by the same uniqueness theorem that gives the Einstein tensor.

9 — Kill Switches

AP08 engages eight kill switches. One (KS-I.6) is closed by AP10 under contingency on AP10's KS-D.3; seven remain live. The closures and live tests target the load-bearing claims of the derivation: the uniqueness theorems of potential theory (KS-I.1) and Lovelock (KS-I.2), the coefficient identification through The Lock (KS-I.3), the structural inaccessibility of κ (KS-I.4), the curvature-record identity (KS-I.5), the dimensional count (KS-I.6), the cosmological constant value (KS-I.7), and the bridge from Axiom R to covariant conservation (KS-I.8).

KS-I.1 — Axiom-to-Poisson chain (via Helmholtz family + small- Λ limit)

Claim. The four axiom-derived symmetry constraints in section 4.1 (Lorentz invariance; locality as differential form via Axiom C; second-order in the metric via QRA; scalar form), applied to the weak-field regime (Constraint 2 as the defining condition of the regime, not an axiom-derived constraint), leave a one-parameter Helmholtz family $a \cdot \nabla^2 \Phi + b \cdot \Phi = A \cdot \rho$ of local, linear, second-order, rotationally-invariant scalar identities. The pure Poisson form $\nabla^2 \Phi = A \rho$ is the $b = 0$ case, recovered as the small-cosmological-constant limit of section 8's full tensor identity, in which the Λ contribution becomes negligible at laboratory scales.

Test. Demonstrate that the four axiom-derived constraints leave a wider or different family than the Helmholtz form. Or: demonstrate that pure Poisson is not the correct small- Λ limit of section 8's equation — that the $b = 0$ case is not the leading-order static weak-field behaviour, or that the corrections at laboratory scales are not negligible under the standard interpretation of Λ . Empirical handle: deviations from inverse-square gravity at laboratory scales that would be inconsistent with pure Poisson behaviour.

Status. LIVE — HARD. The Helmholtz-family identification is the load-bearing reformulation; the small- Λ recovery of pure Poisson is the load-bearing reduction. Both are tested by this kill switch.

Recovery. If KS-I.1 fires at the family-identification level (the constraints leave a different or wider family), the weak-field derivation falls back to “the

constraints leave [the correct family], and pure Poisson is recovered as a member of that family if the corresponding limit holds.” If KS-I.1 fires at the small- Λ -limit level (pure Poisson is not the correct limit), the weak-field expression must be re-derived; the tensor derivation (section 8) is independent of this and survives.

KS-I.2 — Lovelock chain

Claim. The five covariant constraints in section 8.1 (general covariance, symmetric rank-2 tensor, divergence-free source via R bridge, second-order in the metric, four-dimensional) plus Lovelock’s 1971 theorem uniquely determine the Einstein tensor plus cosmological constant as the only symmetric, divergence-free, rank-2 tensor in 3+1 dimensions built from the metric and its first and second derivatives.

Test. Demonstrate an alternative second-order, symmetric, divergence-free tensor in four dimensions that satisfies the constraints. Or: demonstrate that Lovelock’s 1971 theorem fails (a mathematical error) or that Lovelock’s 1972 3+1 uniqueness result does not apply under the constraints.

Status. LIVE — HARD. Lovelock 1971 (general N) and Lovelock 1972 (3+1 uniqueness) are the load-bearing external mathematical claims.

Recovery. If KS-I.2 fires at the mathematical level (Lovelock’s theorem itself fails), the derivation fails at section 8.3 and the form of Einstein’s field equations becomes an identification rather than a derived theorem. If KS-I.2 fires at the application level (the 3+1 uniqueness does not apply as invoked), the recovery path is to re-derive the tensor form under the correct theorem statement. The weak-field Poisson derivation (section 4) is independent and survives.

KS-I.3 — Coefficient identification

Claim. $G = 2\kappa/m_e^2$ (equation 5.1), and therefore $A = 4\pi G = 8\pi\kappa/m_e^2$ (equation 5.2). The identification rests on The Lock’s structural claim that the unique ε of Axiom B is the same ε that appears empirically as the electron, with m_e its mass.

Test. Demonstrate that $G \neq 2\kappa/m_e$ — that The Lock identification of ε with the electron is structurally wrong, or that the relabelling produces a numerical inconsistency. Specific structural handles: re-examine the Lock argument (The Lock, Edition 04); check whether ε is correctly identified with the electron rather than another fundamental particle.

Status. LIVE — HARD. The Lock identification is the load-bearing structural claim.

Recovery. If KS-I.3 fires, the coefficient identification in section 5 fails. The form of equation (4.1) ($\nabla^2\Phi = A\rho$ with A a constant) survives as a derived theorem; only the identification $A = 4\pi G = 8\pi\kappa/m_e$ falls. The Hawking-temperature derivation in section 6.3, which substitutes $G = 2\kappa/m_e$ into the surface-gravity formula, would similarly need to be re-expressed in whichever quantities replace κ and m_e .

KS-I.4 — κ inaccessibility

Claim. κ is structurally inaccessible to independent measurement: the only access to κ is through its consequences (G and m_e), which are the residuals of the break. The structural inaccessibility is not a measurement limitation but a structural consequence of κ being the holding limit of the pre-state, which is the state prior to records and measurement.

Test. Identify an observable that constrains κ without going through G — a structural argument that κ is independently testable, or an empirical handle that pins κ via a different chain. If κ can be measured independently, the structural-inaccessibility claim of section 5.2 is falsified.

Status. LIVE — HARD. The claim is structural rather than empirical, and the failure mode would itself be structural (an alternative chain to κ being identified).

Recovery. If KS-I.4 fires, κ becomes independently testable; the structural circularity of section 5.1 (κ defined as $Gm_e/2$) becomes an empirical question rather than a structural disclosure. The form of Einstein's field equations survives; only the claim that κ cannot be independently measured falls.

KS-I.5 — Curvature-record identity

Claim. Gravity sources from accumulated records: the curvature of the condensate (Mode 0) is determined by the record density (sections 3–4) and ultimately by the stress-energy tensor (section 8). The identity $\rho(x) = n(x) \cdot \langle E \rangle / c^2$ (equation 3.2) holds: matter density and record density are two measurements of the same quantity.

Test. Demonstrate that gravity sources from a quantity structurally distinct from accumulated records. Specific handles: matter without record-history (gravitational sources that do not trace back to coupling events) or records without gravitational signature (coupling events that do not produce mass-energy and therefore do not source curvature).

Status. LIVE — EMPIRICAL. The identity is empirically testable in principle (does all matter source gravity in proportion to its accumulated record content?) but no observed deviation has been reported.

Recovery. If KS-I.5 fires, the identification $\rho = n \cdot \langle E \rangle / c^2$ fails; the matter density and the record density are not the same quantity, and the curvature-record identity sections 3–8 fails at section 3.2. The Poisson and Lovelock derivations survive as formal results about whatever the source quantity is; only the identification of that quantity with accumulated records falls.

KS-I.6 — $N = 3$ (closed by AP10)

Claim. The manifold has dimension $3+1$ (three spatial dimensions plus one temporal direction). Lovelock’s theorem applies in four dimensions, and in four dimensions the unique divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$.

Test. Demonstrate that N differs from 3 (the manifold has 2, 4, or some other number of spatial dimensions). Empirical handles: high-energy physics signatures sensitive to extra dimensions, cosmological observations sensitive to dimensional count. Structural handles: AP10’s argument fails (KS-D.3 fires, in which case the count changes accordingly).

Status. CLOSED by AP10 — contingent on AP10’s KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. With $N = 3$ derived in AP10 under that contingency, the dimensional precondition for Lovelock’s $3+1$

uniqueness is met. If AP10's KS-D.3 fires at any axiom, the count changes, and Lovelock's 3+1 uniqueness theorem may not apply; KS-I.6 reopens in that case.

Recovery. If KS-I.6 reopens (via AP10's KS-D.3 firing), the Lovelock derivation in section 8 fails at Constraint 5. The tensor form would need to be re-derived under whatever Lovelock-theorem statement applies in the corrected number of dimensions. The weak-field Poisson derivation (section 4) is independent of the dimensional count (it explicitly invokes the Laplacian on $\mathbb{R}^3$) and would also need to be re-examined under the corrected count.

KS-I.7 — Cosmological constant value

Claim. The existence of the cosmological constant Λ is predicted by Lovelock's theorem (it appears as a free constant in the unique tensor $a \cdot G_{\mu\nu} + b \cdot g_{\mu\nu}$; in equation 8.1, $b = \Lambda$). The value of Λ is not determined by the derivation; whether it can be computed from a downstream architectural principle is open work.

Test. Measure Λ to a value parametrically incompatible with the architecture's structural constraints (wrong sign, wrong magnitude class, or a value that contradicts the existence prediction). Or: derive Λ 's value from an independent body of work chain and demonstrate that the derivation contradicts the observed value.

Status. LIVE — EMPIRICAL. The existence is predicted; the value is open. Current observations are consistent with the existence prediction (a small positive Λ is measured cosmologically).

Recovery. If KS-I.7 fires at the existence level (no cosmological constant is found, or it is structurally impossible), the form of Lovelock's theorem in section 8.2 still admits the $b \cdot g_{\mu\nu}$ term but with $b = 0$; the derivation's reach is bounded but the form of Einstein's field equations survives. If KS-I.7 fires at the value level (Λ is measured to a structurally impossible value), the derivation requires additional architectural input to compute Λ and the existence prediction holds as a lower bound on what the architecture explains.

KS-I.8 — R → conservation bridge

Claim. Axiom R (record monotonicity) implies, in the continuum limit, the covariant local conservation law $\nabla^\mu T_{\mu\nu} = 0$ used as Constraint 3 of section 8.1. The argument is: records cannot be created or destroyed locally, so the total number of records in any region changes only due to flow across the boundary; in the continuum limit this conservation law translates to the vanishing of the covariant divergence of the stress-energy tensor.

Test. Demonstrate that the map from discrete record monotonicity to covariant local conservation does not hold in the continuum limit — a structural argument that R is compatible with non-zero $\nabla^\mu T_{\mu\nu}$, or that the bridge involves an additional unstated assumption that is itself structurally false.

Status. LIVE — HARD. The bridge is acknowledged as a bridge step rather than a completed derivation in section 8.1's Constraint 3.

Recovery. If KS-I.8 fires, the tensor form derivation fails at Constraint 3 of section 8.1. The Lovelock derivation requires the source tensor to be divergence-free; without that, the Einstein tensor as the unique geometric counterpart is not forced. The weak-field Poisson derivation (section 4) does not invoke covariant conservation and survives. Recovery path: derive the covariant conservation law from R explicitly (closing the bridge), or accept that the tensor form is an identification rather than a forced derivation.

Why these eight

The eight kill switches above cover the load-bearing claims of the paper. Each test addresses a structurally distinct concern.

KS-I.1 and KS-I.2 test the two uniqueness theorems on which the form of the identity rests — the weak-field Poisson form (potential theory) and the tensor form (Lovelock). These are the central derivational steps.

KS-I.3 and KS-I.4 test the coefficient identification — the relabelling $G = 2\kappa/m_e^2$ (depending on The Lock) and the structural inaccessibility of κ (depending on the pre-state status of the holding limit). The form survives a failure here; only the identification of the coefficient falls.

KS-I.5 tests the identification at the source side: gravity sources from accumulated records. This is the empirical falsification handle.

KS-I.6 is the dimensional precondition for Lovelock's 3+1 uniqueness theorem, closed by AP10 contingent on KS-D.3.

KS-I.7 tests the cosmological constant: existence predicted, value open. The existence is structurally forced; the value is empirical/cosmological.

KS-I.8 tests the bridge from Axiom R to the covariant conservation law used as Constraint 3 of section 8. The bridge is honestly acknowledged as not yet a completed derivation.

Two further structural claims that some readings would expect to carry their own kill switch are absorbed into the kill switches above. The Embedding Hypothesis (EH) and the Quantum-Record Alignment (QRA) are both proven in AP20 (KS-7, KS-P.4); any failure would falsify AP20, not AP08 specifically, and AP08's argument would then be untestable on the same manifold. Both are upstream of AP08 and tracked at their source paper; AP08 inherits the closures without re-engaging them.

10 — What This Establishes and What Remains Open

10.1 — What this paper closes

One kill switch directly (KS-I.6, by inheritance from AP10), the form of the weak-field gravitational equation (the Poisson equation as a derived theorem), the form of the full tensor field equations (Einstein with cosmological constant via Lovelock 1971/1972), the strong-field consistency check (Hawking temperature from the condensate picture plus AP06 Theorem 3.1), and the structural identification of the gravitational coupling as a ratio of pre-state holding limit to electron mass squared.

Specifically:

First. The form of the gravitational identity in the static weak-field regime is recovered as the small-cosmological-constant limit of the full tensor identity. Four axiom-derived constraints (Lorentz invariance, locality as differential form, second-order, scalar) applied to the weak-field regime leave the Helmholtz family $a \cdot \nabla^2 \Phi + b \cdot \Phi = A_p$ (equation 4.1); pure Poisson $\nabla^2 \Phi = A_p$ (equation 4.2) is the $b = 0$ case, valid when corrections of order $\Lambda/(\text{laboratory scale})^2$ are negligible.

Second. The form of the gravitational identity in the full tensor case is $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ (equation 8.1), forced by five covariant constraints (general covariance, symmetric rank-2 tensor, divergence-free source via R bridge, second-order in the metric, four-dimensional) plus Lovelock's 1971/1972 theorem. With AP10's closure of $N = 3$ (contingent on AP10's KS-D.3 remaining unfired), the derivation is unconditional in the dimensional sense — unconditional relative to the dimensional precondition, with the standing contingency on KS-D.3 named explicitly throughout.

Third. The cosmological constant Λ is predicted to exist by Lovelock's theorem as a free constant in the unique divergence-free symmetric rank-2 tensor. Λ is not added by hand; its existence is a theorem. Its value is not determined by AP08's derivation (KS-I.7).

Fourth. The coefficient is identified (not derived) via The Lock as $A = 4\pi G = 8\pi\kappa/m_e^2$ (equation 5.2). The structural circularity is honestly disclosed in

section 5.1: κ is defined as $Gm_e^2/2$, so the relabelling is a tautology at the level of numerical computation. What is derived is the *form* of the identity; what is identified is the coefficient.

Fifth. The structural inaccessibility of κ is named: κ is the holding limit of the pre-state, accessible only through its products (G and m_e), because measurement itself is a product of the break that κ describes. Tracked by KS-I.4 as a live structural test.

Sixth. The strong-field limit is consistent with established physics. The Hawking temperature $T_H = \hbar c^3 m_e^2 / (16\pi k_B \kappa M)$ (equation 6.3) is recovered from the Schwarzschild solution of section 8's field equations and standard quantum field theory in curved spacetime (Hawking 1975), with the condensate picture supplying the structural interpretation and AP06 Theorem 3.1 supplying the structural guarantee of nonzero leakage. The numerical formula is not new; the 420 Code contribution is the structural picture under which it is recovered. The m_e in T_H is the same Lock relabelling from section 5.1 reappearing at the horizon scale, viewed there as a substitutional consequence of the coupling identification.

Seventh. The Newton-Einstein relationship is structurally named: Newton's gravity is the linearisation near the common point of the eye topology; Einstein's gravity is the full curve. The architecture derives both. They are the same structure read at different distances from the common point.

10.2 — What this paper does not close

Seven kill switches remain live.

KS-I.1 (Axiom-to-Poisson chain, LIVE — HARD). Structural test on the uniqueness theorem of potential theory under the five constraints of section 4.1.

KS-I.2 (Lovelock chain, LIVE — HARD). Structural test on Lovelock's 1971/1972 theorems under the five constraints of section 8.1.

KS-I.3 (Coefficient identification, LIVE — HARD). Structural test on The Lock identification (ε = electron).

KS-I.4 (κ inaccessibility, LIVE — HARD). Structural test on whether κ can be independently measured.

KS-I.5 (Curvature-record identity, LIVE — EMPIRICAL). Empirical test on whether gravity sources from accumulated records.

KS-I.7 (Cosmological constant value, LIVE — EMPIRICAL). Open: Λ 's existence is predicted, value is not determined.

KS-I.8 ($R \rightarrow$ conservation bridge, LIVE — HARD). The bridge from discrete record monotonicity to covariant local conservation is acknowledged as a bridge step, not a completed derivation.

Full statements at section 9.

10.3 — Items held open without claiming debt status

Two items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as the items tracked by KS-I.3 (The Lock) or KS-I.8 (R bridge).

The cosmological constant value (KS-I.7). Existence predicted by Lovelock. Value is downstream cosmological work — held open at section 0.7 without claiming a separate structural debt because AP08 establishes the existence claim; the value is downstream work.

Constraint 2 of section 4 (the weak-field regime as defining condition). The weak-field regime is what section 4 derives in; the linearity that goes into the Helmholtz family is the defining condition of that regime, not a constraint derived from Axiom R alone. The full nonlinear extension (sections 7–8) does not require this condition. Held open at section 0.7 without claiming a separate debt, because “weak-field” is a chosen regime, not an unresolved structural item.

10.4 — Where the philosophical-register implementations live

AP08 is a structural derivation. Its philosophical-register companions — the chapters where the same structural facts are inhabited in the prose register of the $\emptyset$ Models catalogue — are named here at the level of expected pairings; final placement is confirmed during the $\emptyset$ Models compilation sweep alongside the Notebook II audit programme.

Dissolutions $\emptyset$ (the chapter on gravity and the structure of curvature) carries the philosophical companion to sections 2–7: the condensate as the ground we stand on, gravity as the geometry of what has happened, the eye topology as the full nonlinear shape, Newton-as-linearisation and Einstein-as-full-curve.

Predictions $\emptyset$ carries the empirical face of the derivation: the unconditional Einstein’s field equations, the existence of the cosmological constant, the Hawking-temperature derivation as a consistency check, and the downstream cosmological consequences.

Resolutions $\emptyset$ (the chapter on the structural inaccessibility of pre-state quantities) carries the philosophical companion to sections 5.1–5.2: the structural circularity of κ , the pre-state as the state before measurement, the impossibility of looking behind the break.

The direction of acknowledgment is two-way. AP08 is the original derivation; the $\emptyset$ Models chapters inhabit the result in the prose register. Final companion-pairing list is fixed at $\emptyset$ Models compilation.

10.5 — Structural relationships to subsequent work

AP08 closes Notebook II. Its results feed the entire downstream gravitational and cosmological programme.

Predictions $\emptyset$ inherits the unconditional Einstein’s field equations. The cosmological constant existence prediction is one of the body of work’s headline empirical-falsifiability handles. Predictions $\emptyset$ also inherits the Hawking-temperature derivation as a strong-field consistency check that anchors the 420 Code against established physics.

AP24 (The Residual) inherits the m_e identification used in sections 5 and 6.3. The Lock’s identification of ε with the electron is upstream of AP08’s coefficient identification.

AP09 (The Break — QM) reads against AP08 via the Mode 0 / Mode 1 distinction at section 2.1: gravity is Mode 0 (the condensate’s curvature); quantum mechanics operates as Mode 1 (the individual record-writing events that continue to occur on the structure EH + QRA extracts from the algebra).

Cosmological models built downstream of AP08 inherit the field equations directly. The full nonlinear structure (section 7) and the eye topology (section

7.1) are the structural framework for the body of work's cosmological programme.

10.6 — Placement within The 420 Code

AP08 is the fourth and closing chapter of Notebook II (Spacetime). The chapter sequence runs AP03 (The Ratio) — AP05 (The Break) — AP10 (The Dimension) — AP08 (this paper). AP03 derives the constants; AP05 walks the chain and reformulates the cosmological identity as junction conditions; AP10 pins the dimensional count and identifies the Hilbert space as the 1:1; AP08 closes with the field equations.

AP08 sits at the structural payoff of Notebook II for a specific reason: it is the paper that converts the four axioms (with EH, QRA, and $N = 3$ all in place) into Einstein's field equations as a theorem. Once AP08 has done its work, Notebook II reads as a closed structural chain from the four axioms to the central equations of gravitational physics.

The bidirectional relationship with AP10 is the architectural centre of the Notebook. AP08's tensor derivation required $N = 3$ as a precondition; AP10 derived $N = 3$ from the four independent axioms; the joint output is Einstein's field equations as an unconditional theorem (modulo the contingency on AP10's KS-D.3). Either paper alone would be partial; together they close the chain.

Outside Notebook II, AP08 feeds Predictions $\emptyset$ directly (the unconditional field equations and the cosmological constant existence prediction), AP24 (via the `m_e` identification used in coefficient and Hawking sections), and the downstream cosmological programme.

10.7 — How The 420 Code reads now

AP08 in place closes Notebook II and changes the reading of The 420 Code in five ways.

First. Einstein's field equations are a theorem of the record algebra in the dimensional sense — unconditional relative to the dimensional precondition, contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. The chain runs axioms $\rightarrow$ Lorentzian manifold (AP10 section 4, fenced at KS-D.4) $\rightarrow$ $N = 3$ (AP10, contingent on KS-D.3) $\rightarrow$

record density on M (sections 2–3) → four axiom-derived symmetry constraints plus the weak-field regime definition (section 4) → Helmholtz family with pure Poisson as the small- Λ limit → five covariant constraints (section 8) plus Lovelock 1971/1972 → $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. No link in this chain is conditional on an empirical input for the form of the field equations.

Second. The cosmological constant has a structural origin. Λ is predicted to exist by the same uniqueness theorem that gives the Einstein tensor. It is not added to the equations by hand; its existence is forced. The value is not determined; that is open work.

Third. The gravitational coupling has a structural reading. $G = 2\kappa/m_e^2$: gravity is the ratio of the pre-state holding limit to the square of what escaped (the electron mass). The structural circularity (κ defined as $Gm_e^2/2$) is honestly disclosed; the form of the equations is derived, the value of the coupling is identified.

Fourth. The Newton-Einstein relationship is no longer two theories standing in tension. Newton's gravity is the linearisation of Einstein's gravity near the common point of the eye topology. The architecture derives both from the same structural starting point. Newton sees the straight line near the origin; Einstein sees the curve.

Fifth. The Lock's identification ($\varepsilon = \text{electron}$) carries through to all scales of the gravitational identity. The electron mass appears in the coupling $G = 2\kappa/m_e^2$ at section 5; the same identification reappears in the horizon expression at section 6.3 as a substitutional consequence of that same coupling. This is consistency with The Lock at the horizon scale, not an independent structural result discovered at the horizon. Section 5.1's disclosed circularity (κ defined as $Gm_e^2/2$) is preserved: the m_e in T_H is the same relabelling reappearing, viewed at the horizon.

10.8 — The closing

You have walked through this paper. You have seen the axioms inherited, the condensate named as the manifold that emerges from accumulated records, the record density identified with matter density under The Lock, the Poisson equation recovered in the small-cosmological-constant limit of section 8's full identity (with the Helmholtz family acknowledged as the broader form admitted by the symmetry constraints alone), the coefficient identified (with the

structural circularity honestly disclosed) as a ratio of pre-state holding limit to electron mass squared, the Hawking temperature recovered as a strong-field consistency check (with the m_e appearance at the horizon named as the same Lock relabelling reappearing), the full nonlinear topology named as the eye with Newton at the common point and Einstein in the curve, and the tensor form forced by five covariant constraints plus Lovelock's theorem to be $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$.

Einstein's field equations are a theorem of the record algebra.

Not because we measured the world and wrote the equations down.

Because the four axioms, the proven embedding, the proven quadratic regularity, the derived dimensional count, and Lovelock's uniqueness theorem in 3+1 leave exactly one local, second-order, divergence-free, symmetric, rank-2 covariant identity between curvature and stress-energy.

The cosmological constant exists because the same uniqueness theorem says so.

The gravitational coupling is the ratio of what stayed (κ , the holding limit) to what escaped, squared (m_e^2).

Newton is the linearisation. Einstein is the curve. The architecture is both.

Notebook II is closed.

You are looking at the gravitational sector of The 420 Code as a single closed chain from axioms to field equations.

Claim Summary

Section 0 (Dependency and Scope). Contract with the reader. Dependencies (AP20 for the canonical axioms and the EH and QRA proofs, with the completeness/minimality obligations fenced at KS-P.1/KS-P.2; the Lorentzian signature via AP10 section 4, fenced at KS-D.4; AP10 for $N = 3$ contingent on KS-D.3; AP06 Theorem 3.1; The Lock, Edition 04; Lovelock 1971/1972 as external), axiom role mapping ($S \rightarrow$ eye topology and condensate framing; $B \rightarrow$ The Lock, the coefficient identification, the horizon leakage; $R \rightarrow$ weak-field linearity consistency, the R-conservation bridge, density non-negativity; $C \rightarrow$ differential-form locality, horizon formation, density-change propagation bound), epistemic status per section, kill switch summary, structural debts (The Lock identification; the $R \rightarrow$ conservation bridge), external mathematical dependencies (Lovelock 1971/1972; potential theory; Schwarzschild + Hawking 1975), items held open (cosmological constant value; Constraint 2 weak-field regime definition), structural relationships.

Section 1 (Axioms Inherited). FOUNDATIONAL. The four canonical axioms $\{S, B, R, C\}$ re-stated from AP20 Paper D. EH and QRA proven (AP20). Lorentzian manifold (M, g) with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$ carried via AP10 section 4 (formal derivation fenced at KS-D.4). Special relativity is a theorem of that signature, not an assumption.

Section 2 (The Condensate). STRUCTURAL FRAMING. The manifold understood as the structure that emerges from accumulated records. Mode 0 (the metric, the geometry, the background condition) and Mode 1 (propagation on the condensate, coupling events). Gravity is Mode 0.

Section 3 (Record Density). DEFINITION + IDENTIFICATION. Record number density $n(x)$ defined (equation 3.1). Identification with matter density $\rho(x) = n(x) \cdot \langle E \rangle / c^2$ (equation 3.2) under The Lock. Properties inherited from axioms: $\rho \geq 0$ from R, causality from C, electron coupling from B.

Section 4 (Weak-Field Form). DERIVATION + LIMIT RECOVERY. Four axiom-derived constraints (Lorentz invariance, locality as differential form, second-order in the metric, scalar) applied to the weak-field regime (Constraint 2 as the defining condition of the regime) leave the Helmholtz family $a \cdot \nabla^2 \Phi + b \cdot \Phi = A \cdot \rho$ (equation 4.1). Pure Poisson $\nabla^2 \Phi = A \cdot \rho$ (equation 4.2) is the $b = 0$ case, recovered as the small-cosmological-constant limit of section 8's full identity in which Λ corrections are negligible at laboratory scales.

Section 5 (Coefficient). IDENTIFICATION + HONEST DISCLOSURE. $G = 2\kappa/m_e^2$ (equation 5.1) via The Lock; $A = 4\pi G = 8\pi\kappa/m_e^2$ (equation 5.2); $\nabla^2\Phi = (8\pi\kappa/m_e^2)\cdot\rho$ (equation 5.3). Structural circularity disclosed: κ defined as $Gm_e^2/2$, so relabelling is tautological at the level of numerical computation. κ structurally inaccessible to independent measurement.

Section 6 (Strong-Field Limit). CONSISTENCY CHECK — RECOVERY. Hawking surface gravity $\kappa_H = c^4/(4GM)$ (equation 6.1) and Hawking temperature $T_H = \hbar c^3/(8\pi\kappa_{\text{BGM}})$ (equation 6.2) recovered from the Schwarzschild solution of section 8 plus standard QFT in curved spacetime (Hawking 1975). Under The Lock substitution: $T_H = \hbar c^3 m_e^2 / (16\pi\kappa_{\text{B}}M)$ (equation 6.3). The m_e at the horizon is the same Lock relabelling reappearing, viewed at the horizon scale, not an independent finding.

Section 7 (Full Nonlinear Structure). STRUCTURAL ARGUMENT. The eye topology: two curves ($\mathcal{L}$ and $\mathcal{P}$ from Axiom S) meeting at the common point (the 1:1 pre-state). Closed topology. Newton is the linearisation near the common point; Einstein is the full curve. The Lock identification carries through both ends as a consequence of the same coupling identification, not as an independent finding at either end.

Section 8 (Tensor Form). DERIVATION. Five covariant constraints (general covariance, symmetric rank-2 tensor, divergence-free source via R bridge, second-order in the metric, four-dimensional from AP10 contingent on KS-D.3) plus Lovelock 1971/1972 force $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ (equation 8.1) as the unique symmetric, divergence-free, rank-2 tensor identity in 3+1 dimensions. Einstein's field equations with cosmological constant, with the $\Lambda g_{\mu\nu}$ term retained as the four-dimensional covariant counterpart of the Helmholtz-family $b\cdot\Phi$ term that the weak-field expression drops in the small- Λ limit.

Section 9 (Kill Switches). FORMAL FALSIFICATION REGISTER. KS-I.1 (axiom-to-Poisson, LIVE — HARD), KS-I.2 (Lovelock chain, LIVE — HARD), KS-I.3 (coefficient identification, LIVE — HARD), KS-I.4 (κ inaccessibility, LIVE — HARD), KS-I.5 (curvature-record identity, LIVE — EMPIRICAL), KS-I.6 ($N = 3$, CLOSED by AP10 contingent on KS-D.3), KS-I.7 (cosmological constant value, LIVE — EMPIRICAL), KS-I.8 ($R \rightarrow$ conservation bridge, LIVE — HARD).

Section 10 (What This Establishes and What Remains Open). SYNTHESIS. Seven closures named (form of Poisson; form of Einstein's field equations;

cosmological constant existence prediction; coefficient identification; κ structural inaccessibility; Hawking temperature consistency check; Newton-Einstein structural relationship). Seven kill switches live. Two items held open without claiming separate debt status. Placement within The 420 Code: fourth and closing chapter of Notebook II.

Conditionality Footer

Dependencies

AP20 (the canonical axioms: formal statement section 3.3; completeness and minimality stated at section 4, fenced at KS-P.1/KS-P.2) — the floor of section 1.

The Lorentzian signature (AP10 section 4, fenced at KS-D.4) — Lorentz invariance in section 4.1; general covariance in section 8.1; special relativity as a theorem.

AP20 (EH and QRA proven) — EH underwrites the entire embedding into the smooth manifold; QRA underwrites the second-order curvature constraint in sections 4 and 8.

AP10 (The Dimension) — $N = 3$ derived from the four independent axioms. CLOSED contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. If KS-D.3 fires, KS-I.6 reopens.

AP06 Theorem 3.1 ($\varepsilon > 0$ whenever c is finite) — the leakage premise for the Hawking-temperature derivation in section 6.2.

The Lock, Edition 04 ($\varepsilon = \text{electron}$) — the identification underwriting the coefficient identification in section 5 and the electron-at-horizon result in section 6.3.

Lovelock 1971 / Lovelock 1972 (external mathematical theorems) — the tensor-uniqueness result in 3+1 dimensions invoked in section 8.2.

Dependents

Predictions $\emptyset$ — the unconditional Einstein's field equations and the cosmological constant existence prediction feed the 420 Code's cosmological programme.

Dissolutions $\emptyset$ — the philosophical-register chapter on gravity inherits the condensate framing, the eye topology, and the Newton-Einstein relationship.

Resolutions $\emptyset$ — the philosophical-register chapter on pre-state inaccessibility inherits the κ -circularity disclosure of section 5.

AP09 (The Break — QM) — the Mode 0 / Mode 1 distinction at section 2.1 reads against AP09's quantum-mechanics derivation.

AP24 (The Residual) — m_e identification is upstream of AP08; AP24 also inherits the horizon-leakage result of section 6.

Kill switches closed by this paper

KS-I.6. $N = 3$. CLOSED by AP10 — contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. With $N = 3$ derived in AP10 under that contingency, the dimensional precondition for Lovelock's 3+1 uniqueness is met. Einstein's field equations are an unconditional theorem of the record algebra in the dimensional sense — unconditional relative to the dimensional precondition, with the standing contingency on AP10's KS-D.3 named explicitly throughout.

Kill switches that remain live

KS-I.1 (Axiom-to-Poisson chain, LIVE — HARD). See section 9.

KS-I.2 (Lovelock chain, LIVE — HARD). See section 9.

KS-I.3 (Coefficient identification, LIVE — HARD). See section 9.

KS-I.4 (κ inaccessibility, LIVE — HARD). See section 9.

KS-I.5 (Curvature-record identity, LIVE — EMPIRICAL). See section 9.

KS-I.7 (Cosmological constant value, LIVE — EMPIRICAL). See section 9.

KS-I.8 ($R \rightarrow$ conservation bridge, LIVE — HARD). See section 9.

Structural debts owed

Two structural-class items are tied to AP08's argument as the 420 Code debts. Named here so the structural exposure is visible at the door.

The Lock identification (ε = electron) — load-bearing for sections 3.2, 5, and 6.3. Tracked by KS-I.3.

The $R \rightarrow \nabla^\mu T_{\mu\nu} = 0$ bridge — Constraint 3 of section 8.1 acknowledged as a bridge step, not a completed derivation. Tracked by KS-I.8.

External mathematical dependencies

Separately from the 420 Code structural debts, AP08 invokes the following external mathematical results as established theorems. Listed for transparency; tracked at structural-test level.

Lovelock 1971 / Lovelock 1972 — tensor uniqueness in general N and 3+1 specifically. Tracked by KS-I.2.

Potential theory in the Helmholtz-family identification — the most general second-order linear scalar operator form. Tracked by KS-I.1.

The Schwarzschild solution and Hawking 1975 — the standard surface-gravity formula and Hawking temperature. Used in section 6 as recovery, not derivation.

Items held open without claiming debt status

The cosmological constant value (KS-I.7). Existence predicted by Lovelock; value is downstream cosmological work.

Constraint 2 of section 4 (linearity in the weak-field limit). Motivated by Axiom R but best understood as a consistency requirement between sections 4 and 8 rather than an independent derivation from R alone.

References

Lovelock’s theorem and the gravitational-physics literature. References are restricted to the external mathematical and physical results that AP08 invokes directly: Lovelock 1971 and 1972 (the uniqueness theorems), Hawking 1975 (the strong-field consistency check), and a small set of formative results on the Schwarzschild solution and horizon junction conditions.

Lovelock, D. (1971). *The Einstein tensor and its generalizations*. Journal of Mathematical Physics 12, 498–501. The general N-dimensional uniqueness theorem for divergence-free symmetric rank-2 tensors built from the metric and its first and second derivatives. The theorem invoked at section 8.2; the load-bearing external claim tracked by KS-I.2.

Lovelock, D. (1972). *The four-dimensionality of space and the Einstein tensor*. Journal of Mathematical Physics 13, 874–876. The special uniqueness result in 3+1 dimensions: the divergence-free symmetric rank-2 tensor in four spacetime dimensions takes the form $a \cdot G_{\mu\nu} + b \cdot g_{\mu\nu}$, with no additional higher-order terms admissible. With AP10’s closure of $N = 3$, this is the form invoked at equation (8.1).

Hawking, S. W. (1975). *Particle creation by black holes*. Communications in Mathematical Physics 43, 199–220. The original derivation of Hawking radiation and the Hawking temperature. The strong-field consistency check at section 6.2 builds on this result; AP08’s derivation arrives at the same temperature via the condensate picture plus AP06 Theorem 3.1.

Schwarzschild, K. (1916). *Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie*. Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften, 189–196. The original spherically symmetric solution of the Einstein field equations. Underlies the surface-gravity formula at equation (6.1).

Internal references to The 420 Code (AP01, AP03, AP05, AP06, AP09, AP10, AP20, AP24, The Lock (Edition 04), Dissolutions Ø, Predictions Ø, Resolutions Ø) are catalogued in the Conditionality Footer above. The 420 Code is published copyleft and free at the420code.org; the body of work links to all referenced internal papers.

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